

QuantAMM V1 Audit Summary

Prepared by Cyfrin

Version 1.2

Lead Auditors

0kage

immeas

December 17, 2024

Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 4

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 5

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for QuantAMM on QuantAMM V1. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 1.2 dated December 17, 2024, was delivered to QuantAMM, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from QuantAMM or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 20 days, the Cyfrin team conducted an audit on the QuantAMM V1 code provided by QuantAMM. The findings consist of 4 High, 7 Medium & 3 Low severity issues with the remainder being informational.

Summary

Project Name	QuantAMM V1
Repository	QuantAMM-V1
Commit	7213401491f6...
Audit Timeline	Nov 18th - Dec 13th, 2024
Methods	Manual Review, Stateless Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	4	4	0
Medium Risk	7	6	1
Low Risk	3	2	1
Informational	5	5	0
Gas Optimizations	0	0	0
Total	19	17	2

6 Audit Scope

Cyfrin conducted an audit of QuantAMM based on the code present in the repository commit hash 7213401.

The following contracts were included in the scope of the audit:

```
pkg/pool-quantamm/contracts/QuantAMMWeightedPoolFactory.sol
pkg/pool-quantamm/contracts/MultiHopOracle.sol
pkg/pool-quantamm/contracts/ChainlinkOracle.sol
pkg/pool-quantamm/contracts/QuantAMMWeightedPool.sol
pkg/pool-quantamm/contracts/UpdateWeightRunner.sol
pkg/pool-quantamm/contracts/QuantAMMBaseAdministration.sol
pkg/pool-quantamm/contracts/QuantAMMStorage.sol
pkg/pool-quantamm/contracts/DaoOperations.sol
pkg/pool-quantamm/contracts/rules/MomentumUpdateRule.sol
pkg/pool-quantamm/contracts/rules/MinimumVarianceUpdateRule.sol
pkg/pool-quantamm/contracts/rules/AntimomentumUpdateRule.sol
pkg/pool-quantamm/contracts/rules/PowerChannelUpdateRule.sol
pkg/pool-quantamm/contracts/rules/UpdateRule.sol
pkg/pool-quantamm/contracts/rules/base/QuantammCovarianceBasedRule.sol
pkg/pool-quantamm/contracts/rules/base/QuantammMathMovingAverage.sol
pkg/pool-quantamm/contracts/rules/base/QuantammMathGuard.sol
pkg/pool-quantamm/contracts/rules/base/QuantammGradientBasedRule.sol
pkg/pool-quantamm/contracts/rules/base/QuantammVarianceBasedRule.sol
pkg/pool-quantamm/contracts/rules/base/QuantammBasedRuleHelpers.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

QuantAMM is the first protocol to utilize Temporal Function Market Making as a trade execution mechanism for on-chain quantitative asset management strategies. Built on Balancer V3, each pool receives periodic updates with new weights derived from on-chain trading strategies. These weights are then linearly adjusted using multipliers until the next update is executed. This process dynamically alters the composition of the pool balances based on the optimal allocation determined by the strategies.

Key components:

1. **QuantAMMWeightedPool:** an extension of the Balancer V3 Weighted Pool that supports up to 8 tokens. Within a single block, it behaves like a standard Balancer V3 Weighted Pool. Between blocks, the weights are adjusted using multipliers provided by the pool's trading strategy.
2. **UpdateWeightRunner:** a singleton contract responsible for managing weight updates for all pools. When called for a specific pool, it calculates the new adjusted weights based on the pool's strategy and updates the pool accordingly.
3. **Rules:** the trading strategies currently supported are *Momentum*, *Anti-Momentum*, *Power Channel*, and *Minimum Variance*. Each rule calculates new weights and multipliers for a pool based on the latest oracle prices. Further reading can be found in the QuantAMM paper.

Portfolio managers deposit into QuantAMM pools to use the strategies, and arbitrageurs use the weight changes to adjust the pool composition towards the strategy's target.

9 Attestation

Cyfrin attests that it performed a security review of the QuantAMM V1 code provided by QuantAMM.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-12-17-cyfrin-quantamm-v1.2.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to QuantAMM has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=ea02c08b9e0db089386d0708505c7d8471a7f6f52bef0cb5b889551e59d470d0
F=2024-12-17-cyfrin-quantamm-v1.2-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.